

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Chapter 3

BASIC TOOLS OF MATHEMATICS AND STATISTICS IN ECONOMICS

BASIC TOOLS OF MATHEMATICS:

Modern economists consider the use and help of fundamental mathematical tools to discuss the relationship of various quantities in economics very fruitful. In view of the importance of such relationships we study the following concepts:-

VARIABLES:

Generally the quantities which change their values are known as “variables”. It can be well defined as under:

“A variable is a magnitude which is measurable and taken of different values.”

According to Prof. Kooros, “A variable is a symbol which during a discussion may assume different values or a set of admissible values.”

E.T. Dowling defines “variable” as under:-

“A variable ranges over a set of possible values within a given problem.”

In normal life Time, Temperature, speed of a vehicle etc. are regarded as variables.

In economics there are many quantities which also change their values e.g. Income, Consumptions, Saving, Investment, Price, demand, supply, utility etc.

In Mathematics some letters of alphabet (x,y,z) are used to represent variables.

But in economics generally first capital letter of the quantity is used for variable e.g. P for price, C for consumption, S for saving, I for investment, u for utility etc.

However, sometimes specific symbol may be used e.g. y for National Income, M for imports, X for exports, π for profit etc.

VALUE OF A VARIABLE:

The number which can represent a variable is known its value. If market price of sugar is Rs.25 /- per K.G. Here value of variable $p = 25/-$

During a specific time period maximum value of a variable is called its upper limit and minimum value as “lower Limit”. The distance between upper and lower limit is known as “Range”. Suppose, price of sugar during the year, 2008 remained between Rs.30 and Rs.24 per K.G. This will be the range of variable “P”. While Rs.30 is the upper limit and Rs.24/- is the lower limit of the variable.

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KINDS OF VARIABLES

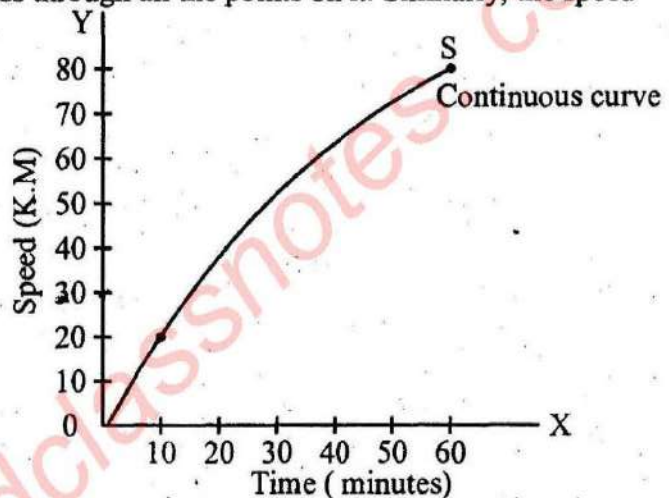
Following are some important kinds of variables.

1. Continuous variable
2. Discontinuous variable
3. Independent variable
4. Dependent variable

1. Continuous Variable:

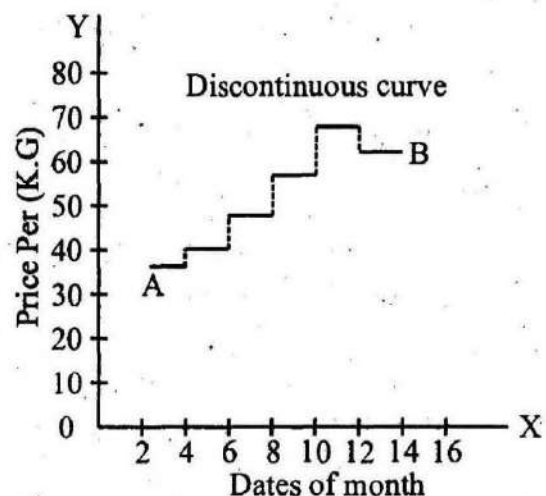
A continuous variable can take on every possible value within its range without leaving any gap or taking any jump. For example, Time is a continuous variable as the needles of a wall clock, or wrist watch pass through all the points on it. Similarly, the speed of a car changes continuously without any jump or interval. If we assume that speed changes from 10 k.m. to 80 k.m. during 60 minutes drive, then the needle of Speedo meter must pass through all possible values between 10 to 80.

In the diagram, it has been shown that "speed" is a continuous variable, which makes a continuous curve "OS" without any gap. It may move upward or downward but there will be no gap or break in it.



2. Dis-continuous Variable:

A discontinuous variable is one which leaves gaps and intervals while changing its values. The curve of a discontinuous variable depends upon various pieces and parts. It is broken between different values of the variables. In the graph price of ghee per K.G. has been shown on different dates of a month. Curve AB consists of various pieces as price increases from Rs.40/- per K.G. to Rs.80/ per K.G. on different dates.



3. Independent vs Dependent Variables:

Some variables have functional relationship with each other. The change in anyone influences the value of the other variable. A variable which can assume any value at its own (independently) is known as an independent variable. When such a change in an independent variable causes a change in the value of another variable that will be regarded as a dependent variable. Thus a dependent variable is the one whose value is determined by the values of an independent variable present in the Model.

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According to the law of demand $q_d = f(p)$. It means that any change in its quantity demanded will depend upon a change in its price (P) thus price (P) of the good will cause a change in its quantity demanded. Here price (P) is an independent variable while quantity demanded (q_d) is a dependent variable. This situation also indicates that demand is a function of price.

Similarly, $C = f(y)$ (consumption function)

And $S = f(y)$ (saving function)

Here y (income) is an independent variable while, c (consumption) and S (saving) are dependent variables.

When relationship of variables is discussed through equations, independent variable is written on R.H.S. such as $q_d = 10 - 2p$. Here variable p is independent while q_d is dependent.

CONSTANTS:

The value of a constant is "known" and "fixed". According to E.T. Dowling, "A constant is a quantity that does not change in a given period".

A numerical constant always keeps the same value during discussion e.g. 2, 4, 10, 50 etc. while a symbolic constant owns a single value for one problem, while for other discussions it may adapt different values e.g. a , b , c etc.

The numbers in any equation are constants. For instance $4x + 2 = 6$. IN this equation 4, 2, 6 are constants while x is a variable. Any value given to x cannot change these numerical constants.

PARAMETERS

There are certain quantities whose values keep on changing in real life but for specific purposes:-

These are assumed to "remain the same." These quantities are known as "parameters."

For instance, quantity supplied of a commodity depends upon many variables e.g. price (p), cost of production (c) Method of production (M) and the price of raw material and machines (P_i). Thus $q_s = f(P, C, M, P_i)$.

But to prove the law of supply true it is assumed that only " P " changes while, C, M, P_i remain constant, which are parameters. Thus the above function may be written as $q_s = a + bp$, Here a, b are parameters. We can say that in Economics, the assumptions of any law are "parameters".

FUNCTION

Some variables are inter-related with each other. When one variable (y), depends upon other variable (x) then $y = f(x)$. Variable " y " is function of variable " x ". expresses a relationship between two variables (x, y) such that for each value of x there exists one and only one value of y .

In economics there are certain variables which have "functional relationship with each other e.g. $c = f(y)$ $q_s = f(p)$ etc.

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FORMS OF FUNCTIONS

1. **Explicit Function:**

When a variable is totally dependent upon the other variable e.g. $y = f(x)$. Then y is an explicit function of x . One assumed value of x will result only one value of y .

Example: (1) $y = 4x + 1$ when $x = 2$ then $y = 4(2) + 1 = 9$. In economics following are explicit functions:

(i) $q_d = f(p)$, $q = 10 - 2p$

(ii) $q_s = f(p)$, $q = 4 + p$

(iii) $c = f(y)$, $c = 10 + 0.8y$

Iv) $s = f(y)$, $s = -10 + 0.2y$

2. **Implicit Function:**

When two variables depend upon each other by giving value to one variable we get the value of the other variable i.e. (x or y), the function is known as an implicit function.

Example: (i) $xy = 10$ (ii) $2x + 4y = 16$ (iii) $y^2 - x^2 = 20$

In economics the relationship between (y) income, and (I) investment represents the implicit function because

(i) $y = f(I)$ (In view of Multiplier)

(ii) $I = f(y)$ (In view of Accelerator)

3. **Inverse function:**

The two explicit functions of one implicit function are inverse functions of each other. For example (i) $xy = 10$ (implicit function)

Explicit functions (a) $x = \frac{10}{y}$ (b) $y = \frac{10}{x}$ a and b are inverse functions of each other.

(ii) $y = f(I)$ we get inverse function $I = f(y)$

4. **Increasing function:**

When the values of independent and dependent variables move in the same direction (rising or falling) that will be regarded an increasing function. If $y = f(x)$ and $y = 10 + 2x$ when value of x rises, value of y also rises and vice versa.

In economics the examples of increasing function are:

(i) $q_s = f(p)$ Supply function $q_s = 4 + p$

(ii) $C = f(Q)$ Cost function $C = 1 + q^2$

(iii) $R = f(Q)$ Revenue function $R = 2 + 3q + q^2$

(iv) $C = f(y)$ Consumption function $C = 50 + 0.8y$

(v) $S = f(y)$ Saving function $S = -50 + 0.2y$

(vi) $I = f(y)$ Investment function $I = 25 + 0.05y$

5. **Decreasing function:**

When the values of independent and dependent variables move in opposite direction. If $y = f(x)$ and an increase in the value of x results a decrease in the value of y and vice versa (negative co-relation) then there exists a decreasing function. If $y = f(x)$ and $y = 12 - 2x$.

Examples in Economics:

(i) $q_d = f(p)$, $q_d = 10 - 2p$ (demand function)

(ii) $M_u = f(Q)$, $M_u = 8 - 3q$ (utility function)

(iii) $I = f(i)$ Investment function w.r.t. interest rate.

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6. Single valued function:

If there is one and only one value of dependent variable in view of one given value to independent variable that is known as a "single value function". A linear equation is a such type of function such as.

$$(i) y = 3x - 2 \quad (ii) qd = 8 - 2p$$

7. MULTI-VALUED FUNCTION:

This type of function indicates more than one values of a dependent variable while placing only one value to an independent variable. For example, $y^2 = x^2$ here $y = \pm \sqrt{x^2}$ as

(i) $x^2 = x(x)$ (ii) $-x(-x)$ thus one value of $x = 2$ will result two values of y

$$y = \pm \sqrt{2^2} = \pm \sqrt{4} = \pm 2 \quad (i) y = 2 \quad (ii) y = -2$$

EQUATIONS

An equation expresses the equality between two algebraic expressions. More simply we can say that "An equation is a statement that two algebraic expressions are equal". The sign of equation is (=). An algebraic expression is a combination of Numbers, letters with + or - signs. Every part with + or - signs is known as "Term". For example (i) $2x + 4 = 4x - 6$ (ii) $x + y = 5$ are equations. In equation No. (i) $2x + 4$ is an algebraic expression having two terms: (i) $2x$ (ii) $+4$. The two algebraic expressions are called L.H.S and R.H.S. (left and Right hand sides) of the equation. One side of the equation may be zero e.g. $4x - 2 = 0$. According to "Terms" algebraic expressions are as under:

- (i) **Binomial:** An algebraic expression with only two terms. e.g. $3x - 5$ or $2x + 6$.
- (ii) **Trinomial:** An algebraic expression with three terms e.g. $x^2 + 4x - 1$
- (iii) **Polynomial:** The algebraic expression with four or more than four terms e.g. $x^3 + 4x^2 - 2x + 8$

SYMBOLS OF EQUATIONS

1. Unknowns:

Un-known are the variables present in the equation in the form of symbols e.g. x, y, z , etc. The equation is to be solved to find out the value of un-known which makes the two sides equal. For instance, in equation (i) $4x + 2 = 0$ there is only one unknown x . but in (ii) $4x + 2y = 16$ x and y are two unknowns.

2. Constants:

These are the numbers or quantities whose values remain un-changed in an-equation e.g. $3x^2 + 4x = 5$ in this equation 3, 4, and 5 are numerical constants.

3. Co-efficient:

Co-efficient is the number or parametric symbol written on the left side of the unknown in an equation. For example (i) $3x - 2 = 0$, Here 3 is co-efficient of x which means that x is added 3 times e.g. $3x = x + x + x$ (ii) $4x^2 + 2x + 10 = 0$. Here $4x^2 = x^2 + x^2 + x^2 + x^2$ and $2x = x + x$. It means that 4 is the co-efficient of x^2 and 2 is co-efficient of x .

4. Index or Power:

The number or symbolic letter written above the un-known on its right side is regarded power of the un-known. It shows the number of times the unknown has been

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multiplied by itself e.g. x^5 shows that 5 is the power of x thus $x^5 = x.x.x.x.x$.

If $x^a - y = 3$ it means that the power of x is a while that of y is 1.

NOTE:

- (i) When the power of the unknown in an equation is positive and real number it is called "Rational equation e.g. $2x^3 + 3x^2 - x = 16$.
- (ii) On the other hand, if the power of unknown is negative and non real number that is called "Irrational equation." e.g. $\bar{x}^3 + 4x^{1/2} + 3 = 0$, $(-3$ and $\frac{1}{2}$ are not positive real numbers)
- (iii) When the power of a variable or constant is zero (0), its value will be equal to to one(1), e.g (i) $x^0 = 1$ (ii) $10^0 = 1$,

FORMS OF EQUATIONS

These can be classified as under:

1. ACCORDING TO THE NUMBER OF VARIABLES

(a) **One Variable equation:**

It is an equation where only one variable has been used. For example,

(i) $4x + 2 = 8$ (ii) $3x - 1 = 0$. In the above equations only one unknown x is present.

(b) **Two variables equation:**

In such an equation two variables (unknown) are used e.g. $3x + 4y + 8 = 0$. In this equation x and y are two unknown.

(c) **Three variables equation"**

Here three variables are used in an equation.

(i) $x + y + z = 20$ (ii) $3x^2 + 2y - 4z = 18$

In the above equations x, y and z are three un-known. Thus we can say that the equation will be classified in view of the number of variables (unknown) used in it.

2. ACCORDING TO POWER OF VARIABLES

(a) **Linear equation**

When in an equation, the maximum power of the unknown (variable) in one, it is known first degree or linear equation. For example (i) $4x + 10 = 0$

(ii) $-4x + 4 = 0$. Here the power of x is 1. The standard form of a linear equation is,

$ax + b = 0$ Here, a and b are constants, where the value of a may be positive (+) or negative (-) but $a \neq 0$ (not zero) While the value of b may be +, -, or 0 (zero).

(b) **Quadratic equation**

If maximum power of unknown is two in an equation. It is known as a "second degree equation" or "quadratic equation".

(i) $x^2 + 2x - 3 = 8$ (ii) $4x^2 - 3x + 10 = 2$. The standard form of quadratic equation is:

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$ax^2 + bx + c = 0$ Here (i) a can have any value (+ or -) except zero ($a \neq 0$) (ii) b and c can have any value (+, -, 0)

Note: [A quadratic equation has always two solutions (roots)]

(c) **Cubic equation**

Maximum power of the variable in such an equation is three. It is also known as "third degree equation; e.g. $x^3 + 2x^2 + 3x - 5 = 0$

Standard form of the cubic equation is:

$$ax^3 + bx^2 + cx + d = 0$$

Here $a \neq 0$, while b, c, d may have any value. Thus the equation may be classified according to the power of unknown it has.

3. SIMULTANEOUS EQUATIONS:

The equations in which number of variables (unknown) is more than one and the solution applied upon then results equality when taken together, are known as "simultaneous equations". If there are two unknown, at least a set of two equations is required simultaneously to find out the solution. These equations may be "linear" or "quadratic" e.g.

(i) $4x + 2y = 16$

(ii) $2x - 4y = 8$

And (i) $2x + y = 4$

(ii) $x^2 + 3y^2 = 10$

Solution of simultaneous equations means to find such values of unknowns which can satisfy and prove true all the equations at the same time.

4. IDENTICAL EQUATIONS

When every value of the unknown present in the equation provides the root of equation which cannot be denied under any circumstances, will be regarded as "identical equation". The sign of such an equation is: $\equiv$

Examples:

(i) $(a + b)^2 \equiv a^2 + b^2 + 2ab$

(ii) $(x+y)(x-y) \equiv x^2 - y^2$

(iii) $x^2 - 4 \equiv (x-2)(x+2)$

Any value given to a, b, x, y will satisfy the identity between both side (L.H.S. $\equiv$ R.H.S)

In economics $S \equiv I$ and $PT \equiv Mv$ are the examples of identical equations.

SOLUTION OF LINEAR EQUATIONS

Example: Solve the equation $5x + 4 = 3x + 12$

Solution: We take the variables along with the co-efficients on L.H.S. and constants on R.H.S.. Thus $5x - 3x = 12 - 4$ (signs are changed during shifting process)

$$2x = 8,$$

$$x = \frac{8}{2} = 4$$

$$x = 4$$

Hence 4 is the root of equation.

Proof or Test:

Putting the value of x (4) on both sides of the equation.

$$5x + 4 = 3x + 12 \quad 5(4) + 4 = 3(4) + 12,$$

$$24 = 24$$

$$\text{L.H.S.} = \text{R.H.S.}$$

Example 2: $4(2x - 5) = 3(2x + 8)$

Solution: $8x - 20 = 6x + 24$

$$8x - 6x = 24 + 20$$

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$$2x = 44, \quad x = \frac{44}{2} = 22$$

Thus

$$x = 22$$

Proof or Test:

Putting the values of $x(22)$ on both sides of the equation $8x - 20 = 6x + 24$

$$8(22) - 20 = 6(22) + 24$$

$$176 - 20 = 132 + 24$$

$$156 = 156$$

$$\text{L.H.S.} \quad \text{R.H.S.}$$

Example 3: $\frac{2x-5}{3} - \frac{3x-1}{4} = \frac{3}{2}$

Solution: Taking the L.C.M $\frac{12(2x-5)}{3} - \frac{12(3x-1)}{4} = \frac{12(3)}{2}$

$$4(2x-5) - 3(3x-1) = 6(3)$$

$$8x - 20 - 9x + 3 = 18$$

Shifting Process $8x - 9x = 18 + 20 - 3$

$$-x = +35$$

$$x = -35$$

Example 4: $\frac{3x-5}{6x+10} = \frac{4x+6}{8x-2}$

Solution: Using cross multiplication method

$$(3x-5)(8x-2) = (4x+6)(6x+10)$$

$$24x^2 - 40x - 6x + 10 = 24x^2 + 36x + 40x + 60$$

$$24x^2 - 46x + 10 = 24x^2 + 76x + 60$$

$$-46x - 76x = 60 - 10$$

$$-122x = 50$$

$$x = -\frac{50}{122}$$

$$x = -\frac{25}{61} \text{ or } -0.409$$

EXERCISE -1

Solve the following equations:

(1) $3x + 5 = 20$

(3) $3x - 12 = 0$

(5) $3(x-2) = 2(3x-1)$

(7) $\frac{x-4}{3} - \frac{x-4}{2} = 1$

(9) $2x + \frac{4-x}{2} = 3x - \frac{8(x-2)}{4}$

(2) $5x - 3 = 2x + 9$

(4) $3x + 17 = 2x + 20$

(6) $3x - 2 = 4x + 3(6x + 8)$

(8) $\frac{3(x+2)}{2} + \frac{2(4x-8)}{5} = \frac{8(2x-6)}{6}$

(10) $\frac{6x-7}{9} = \frac{2(3x-1)}{5}$

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$$\begin{aligned} (11) \quad \frac{1-8x}{4+3x} &= \frac{2+16x}{3-6x} & (12) \quad \frac{2x-4}{5x+2} &= \frac{4x-8}{10x+3} \\ (13) \quad \frac{7x-1}{4} - \frac{1}{3} \left(2x - \frac{1-x}{2} \right) &= \frac{19}{3} & (14) \quad \frac{5x-2}{3x+1} &= \frac{5x-1}{3x+2} \\ (15) \quad \frac{9x-7}{6x+5} &= \frac{6x-3}{4x+2} \end{aligned}$$

ANSWERS

$$\begin{aligned} (1) \quad x &= 5 & (2) \quad x &= 4 & (3) \quad x &= 4 & (4) \quad x &= 3 \\ (5) \quad x &= -1\frac{1}{3} & (6) \quad x &= -1\frac{7}{19} & (7) \quad x &= -2 & (8) \quad x &= -18 \\ (9) \quad x &= 4 & (10) \quad x &= \frac{-17}{24} & (11) \quad x &= \frac{-1}{20} & (12) \quad x &= 2 \\ (13) \quad x &= 7 & (14) \quad x &= 1\frac{1}{2} & (15) \quad x &= \frac{1}{22} \end{aligned}$$

SOLUTION OF QUADRATIC EQUATIONS

Two methods are used to solve a quadratic equation

1. Factorization Method
2. Formula Method

1. FACTORIZATION METHOD:

Under this method following steps are taken.

- (a) The equation is arranged in standard form. $ax^2 + bx + c = 0$
- (b) a is multiplied by c ($a.c$). It is the product of $a.c$
- (c) The two factors of $a.c$ (product) are found in such a way that:
 - (i) Sum total of a and c ($a+c$) = b
 - (ii) The product of factors = $b = (a.c)$

Example 1: $2x^2 + 3x - 5 = 0$

Solution: It is already in standard form

$$a = 2, b = 3, c = -5$$

$$\text{Product of } ac = 2(-5) = -10$$

$$\text{Factors of } -10 = b_1 + b_2 = 3 \text{ and product } (b_1 b_2) = -10$$

Such factors are -2 and 5

Now we write the equation as under

$$\begin{aligned} 2x^2 + 3x - 5 &= 0 \\ 2x^2 + (-2+5)x - 5 &= 0, \quad 2x^2 - 2x + 5x - 5 = 0 \\ 2x(x-1) + 5(x-1) &= 0 \\ (x-1)(2x+5) &= 0 \end{aligned}$$

Now when $x-1=0$

$$x = 1$$

$$2x+5=0$$

$$2x = -5$$

$$x = \frac{-5}{2} = -2.5$$

The two solutions (roots) are (i) $x = 1$ (ii) $x = -2.5$

Proof / Test: $2x^2 + 3x - 5 = 0$

Putting $x=1$, $2(1)^2 + 3(1) - 5 = 0$

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$$\begin{aligned} 2 + 3 - 5 &= 0 \\ 5 - 5 &= 0 \\ 0 &= 0 \end{aligned}$$

Putting $x = -2\frac{1}{2}$ or $\left(-\frac{5}{2}\right)$

$$2\left(-\frac{5}{2}\right)^2 + 3\left(-\frac{5}{2}\right) - 5 = 0$$

$$2\left(\frac{25}{4}\right) - \frac{15}{2} - 5 = 0$$

$$\frac{25}{2} - \frac{15}{2} - 5 = 0$$

$$\frac{25 - 15 - 10}{2} = 0$$

$$25 - 25 = 0$$

$$0 = 0$$

Thus $x = 1$, $x = -\frac{5}{2}$ are the true values of x in the above equation.

Example 2: $2x^2 + 11x + 15 = 0$, Here $a = 2$, $b = 11$, $c = 15$

Solution: $ac = 30$ Factors are 6 and 5.

Putting the values of b (6, 5) in place of 11

$$2x^2 + 11x + 15b = 0$$

$$2x^2 + 6x + 5x + 15 = 0$$

$$2x(x + 3) + 5(x + 3) = 0$$

$$2x + 5 = 0$$

$$2x = -5$$

$$x = \frac{-5}{2} = -2.5$$

$$x + 3 = 0$$

$$x = -3$$

$$x = -3$$

$$x = -2.5$$

Thus

Proof / Test: Putting the values of $x = -2.5$

$$2x^2 + 11x + 15 = 0$$

$$2(-2.5)^2 + 11(-2.5) + 15 = 0$$

$$2(6.25) - 27.5 + 15 = 0$$

$$12.5 - 27.5 + 15 = 0$$

$$27.5 - 27.5 = 0$$

$$0 = 0$$

Now at $x = -3$

$$2x^2 + 11x + 15 = 0$$

$$2(-3)^2 + 11(-3) + 15 = 0$$

$$2(9) - 33 + 15 = 0$$

$$18 - 33 + 15 = 0$$

$$33 - 33 = 0$$

$$0 = 0$$

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Example 3: $2x^2 + x - 1 = 0$ Here $a = 2, b = 1, c = -1$

Taking a.c and making factors

$$\begin{aligned} 2x^2 + x - 1 &= 0 \\ 2x^2 + 2x - x - 1 &= 0 \\ 2x(x + 1) - 1(x + 1) &= 0 \\ (2x - 1)(x + 1) &= 0 \\ 2x - 1 &= 0 \\ 2x &= 1 \\ x &= \frac{1}{2} \\ x + 1 &= 0 \\ x &= -1 \end{aligned}$$

Thus $x = -1$ and $x = \frac{1}{2}$

Proof / Test: When $x = -1$

$$\begin{aligned} 2x^2 + x - 1 &= 0 \\ 2(-1)^2 + (-1) - 1 &= 0 \\ 2 - 1 - 1 &= 0 \\ 2 - 2 &= 0 \\ 0 &= 0 \end{aligned}$$

When $x = \frac{1}{2}$

$$\begin{aligned} 2x^2 + x - 1 &= 0 \\ 2\left(\frac{1}{2}\right)^2 + \frac{1}{2} - 1 &= 0 \\ 2\left(\frac{1}{4}\right) + \frac{1}{2} - 1 &= 0 \\ \frac{1}{2} + \frac{1}{2} - 1 &= 0 \\ 1 - 1 &= 0 \\ 0 &= 0 \end{aligned}$$

Example 4: $5x^2 - 4x = 1$

Keeping in standard form

$$\begin{aligned} 5x^2 - 4x - 1 &= 0 \\ 5x^2 - 5x + x - 1 &= 0 \\ 5x(x - 1) + 1(x - 1) &= 0 \\ (5x + 1)(x - 1) &= 0 \\ x - 1 &= 0, x = 1 \\ 5x + 1 &= 0 \\ 5x &= -1 \\ x &= -\frac{1}{5} \end{aligned}$$

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Proof / Test:

When

$$\begin{aligned}x &= 1 \\5x^2 - 4x &= 1 \\5(1)^2 - 4(1) &= 1 \\5 - 4 &= 1 \\1 &= 1\end{aligned}$$

When

$$\begin{aligned}x &= -\frac{1}{5} \\5x^2 - 4x &= 1 \\5\left(-\frac{1}{5}\right)^2 - 4\left(-\frac{1}{5}\right) &= 1 \\5\left(\frac{1}{25}\right) + \frac{4}{5} &= 1 \\\frac{1}{5} + \frac{4}{5} &= 1 \\\frac{1+4}{5} &= 1 \\\frac{5}{5} &= 1 \\1 &= 1\end{aligned}$$

EXERCISE 2

Solve with the help of Factorizing Method:

- | | |
|-------------------------|--------------------------|
| (1) $x^2 - 2x = 3$ | (2) $x^2 - 4x = 32$ |
| (3) $x^2 - 4x + 3 = 0$ | (4) $x^2 + 3x = 4$ |
| (5) $3x^2 + 5x - 2 = 0$ | (6) $4x^2 - 11x + 6 = 0$ |
| (7) $x^2 - 5x + 6 = 0$ | (8) $x^2 + x - 6 = 0$ |
| (9) $9x^2 - 3x - 6 = 0$ | (10) $x^2 - 7x + 12 = 0$ |

ANSWERS

- | | | |
|-----------------|---------------------------|---------------------------|
| (1) $x = 3, -1$ | (2) $x = 8, -4$ | (3) $x = 3, 1$ |
| (4) $x = 1, -4$ | (5) $x = -2, \frac{1}{3}$ | (6) $x = 2, \frac{3}{4}$ |
| (7) $x = 2, 3$ | (8) $x = -3, 2$ | (9) $x = 1, -\frac{2}{3}$ |
| (10) $x = 3, 4$ | | |

2. FORMULA METHOD

A quadratic equation may be solved with the help of following formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Example 1: $x^2 + 3x = 1$

Arranging the above equation into standard form.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

$$x^2 + 3x - 1 = 0 \quad \text{Thus } a=1, b=3, c=-1$$

Putting these values into formula

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-3 \pm \sqrt{(3)^2 - 4(1)(-1)}}{2(1)} \\ x &= \frac{-3 \pm \sqrt{9+4}}{2} \\ &= \frac{-3 \pm \sqrt{13}}{2} \\ &= \frac{-3 + 3.6}{2}, \frac{-3 - 3.6}{2} = \\ &= \frac{.6}{2} = .3, \frac{-6.6}{2} = -3.3 \\ x &= -3.3 \text{ and } x = .3 \end{aligned}$$

Thus

Example 2:

$$x^2 - 2x = 3$$

$$x^2 - 2x - 3 = 0 \text{ (Standard form)}$$

$$a=1, b=-2, c=-3$$

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 1 \times -3}}{2 \times 1} \\ &= \frac{2 \pm \sqrt{4+12}}{2} \\ &= \frac{2 \pm \sqrt{16}}{2} \\ &= \frac{2 \pm 4}{2} \\ &= \frac{2+4}{2}, \frac{2-4}{2} \\ &= \frac{6}{2} = 3, \frac{-2}{2} = -1 \\ x &= 3, x = -1 \end{aligned}$$

EXERCISE - 3

Solve the following with the help of formula:

1. $3x^2 + 4x - 5 = 0$

2. $4x^2 - x - 3 = 0$

3. $x^2 + 12x + 35 = 0$

4. $x^2 + 21x = 46$

5. $4x^2 + 12x = 7$

Answers:

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

1. $x = .78, x = -2.12$
2. $x = 1, x = -.75$
3. $x = -5, x = -7$
4. $x = 2, x = -23$
5. $x = 5, x = -3.5$

SOLUTION OF SIMULTANEOUS EQUATION

1. LINEAR EQUATION

Two Methods are used:

1. Substitution Method
2. Elimination Method

1. SUBSTITUTION METHOD:

To use this method, we find value of one unknown from the equation and put this value in the other equation.

Example 1:

(i) $5x - 3y = -5$

(ii) $2x + y = 9$

We find the value of y in equation (ii)

$$2x + y = 9$$

$$y = 9 - 2x$$

Putting $y = 9 - 2x$ in equation (i)

$$\begin{aligned} \text{(i)} \quad 5x - 3y &= -5 \\ 5x - 3(9 - 2x) &= -5 \\ 5x - 27 + 6x &= -5 \\ 5x + 6x &= 27 - 5 \\ 11x &= 22 \\ x &= 2 \end{aligned}$$

Now we put the value of $x = 2$ in equation (i) and find the value of y .

$$\begin{aligned} 5(2) - 3y &= -5 \\ 10 - 3y &= -5 \\ -3y &= -15 \\ 3y &= 15 \\ y &= 5 \end{aligned}$$

In both equations when $x = 2, y = 5$

Putting the value of $x = 2$ in equation (ii)

$$\begin{aligned} 2x + y &= 9 \\ 2(2) + y &= 9 \\ 4 + y &= 9 \\ y &= 9 - 4 \\ y &= 5 \end{aligned}$$

Example 2:

(i) $3x - 2y = 7$ (ii) $2x + 5y = 11$

we find the value of x w.r.t. y .

$$\begin{aligned} 3x - 2y &= 7 \\ 3x &= 7 + 2y \end{aligned}$$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

$$x = \frac{7+2y}{3}$$

Putting the value of x in equation (ii)

$$2\left(\frac{7}{3} + \frac{2}{3}y\right) + 5y = 11$$

$$\frac{14}{3} + \frac{4}{3}y + 5y = 11$$

Putting the value of y = 1 in equation (i)

$$3x - 2(1) = 7$$

$$3x - 2 = 7$$

$$3x = 7 + 2$$

$$x = 3$$

$$X = 3, y = 1$$

$$\frac{14 + 4y + 15y}{3} = 11$$

$$14 + 4y + 15y = 33$$

$$14 + 19y = 33$$

$$19y = 33 - 14$$

$$19y = 19$$

$$Y = \frac{19}{19} = 1$$

2. ELIMINATION METHOD

We eliminate any unknown by equating co-efficients in both the equations.

Example 1:

(i) $2x + 6y = 20$

(ii) $x - 2y = 5$

To equate co-efficients of x, we multiply equation (i) by 2

(i) $2x + 6y = 20$

(ii) $2x - 4y = 10$

Subtracting equation (ii) from (i)

(i) $2x + 6y = 20$

(ii) $-2x - 4y = -10$

$$0 + 10y = 10$$

$$10y = 10$$

$$y = 1$$

Putting this value of y (1) in equation (ii)

$$x - 2y = 5$$

$$x - 2(1) = 5$$

$$x - 2 = 5$$

$$x = 5 + 2$$

$$x = 7$$

Thus the solution is y=1, x = 7

Example 2:

(i) $4x + 3y = 7$

(ii) $2x - 2y = 9$

We multiply equation (ii) by 2

$$4x + 3y = 7$$

$$-4x + 4y = -18$$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

$$7y = -11$$
$$y = -\frac{11}{7} = -1\frac{4}{7}$$

Putting the value of $y = -\frac{11}{7}$ in equation (i)

$$(i) \quad 4x + 3\left(-\frac{11}{7}\right) = 7$$

$$4x - \frac{33}{7} = 7$$

$$4x = 7 + \frac{33}{7}$$

Multiplying both sides by 7

$$28x = 49 + 33$$

$$28x = 82$$

$$x = \frac{82}{28} = \frac{41}{14}$$

Thus $x = 2\frac{13}{14}$ and $y = -1\frac{4}{7}$

EXERCISE - 4

Solve the following simultaneous equations and test also.

- | | |
|---------------------|-----------------|
| (1) $2x - y = 3$ | $x + 2y = 14$ |
| (2) $3x - 2y = 7$ | $x + 2y = 5$ |
| (3) $3x + 4y = 7$ | $2x + y = 3$ |
| (4) $5x + 4y = 220$ | $4x + 5y = 230$ |
| (5) $4x + 5y = 6$ | $3x - 2y = 10$ |

ANSWERS:

- | | |
|------------------------|--------------------|
| (1) $x=4, y=5$ | (2) $x=3, y=1$ |
| (3) $x=1, y=1$ | (4) $x=20, y=30$ |
| (5) $x=2\frac{16}{23}$ | $y=-\frac{22}{23}$ |

Solution of simultaneous equations when one is linear and the other is quadratic.

Solution:

1. Find out the value of one unknown in linear equation.
2. Put this value in quadratic equation.
3. Arrange it in standard form.
4. Apply Factorization or Formula Method.
5. Two values of x and two values of y is the solution of these equations.

Example 1:

- (i) $x + y = 7$
(ii) $3x^2 + xy - y^2 = 81$

We find the value of y w.r.t. value of x in equation (i)

- (i) $x + y = 7$
(ii) $y = 7 - x$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Putting the value of y in equation (ii)

$$(ii) \quad 3x^2 + x(7-x) - (7-x)^2 = 81$$

$$3x^2 + 7x - x^2 - (49 - 14x + x^2) = 81$$

$$3x^2 + 7x - x^2 - 49 + 14x - x^2 - 81 = 0$$

$$3x^2 - 2x^2 + 21x - 130 = 0$$

$$x^2 + 21x - 130 = 0$$

This is the standard form of quadratic equation.

$$a = 1, b = 21, c = -130$$

FORMULA METHOD

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-21 \pm \sqrt{(21)^2 - 4(1)(-130)}}{2 \times 1}$$

$$= \frac{-21 \pm \sqrt{441 + 520}}{2}$$

$$= \frac{-21 \pm \sqrt{961}}{2}$$

$$= \frac{-21 \pm 31}{2}$$

$$= \frac{-21 - 31}{2}, \quad \frac{-21 + 31}{2}$$

$$= \frac{-52}{2} = -26, \quad \frac{10}{2} = 5$$

Thus

$$x = -26, \quad x = 5$$

Putting value of x = 5 in equation (i)

$$x + y = 7$$

$$y = 7 - x$$

$$y = 7 - 5 = 2 \quad y = 2$$

Putting value of x = -26 in equation (i)

$$\text{Thus } y = 7 - (-26) = 7 + 26 = 33$$

$$x = 5, \quad y = 2$$

$$x = -26, \quad y = 33$$

Example 2:

$$(i) \quad 4xy = -30$$

$$(ii) \quad 5x + 4y = 5$$

We find the value of y in equation (ii)

$$(ii) \quad 5x + 4y = 5$$

$$4y = 5 - 5x$$

$$y = \frac{5 - 5x}{4}$$

Putting the value of y in equation (i)

$$(i) \quad 4xy = -30$$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

$$4x \left(\frac{5-5x}{4} \right) = -30$$

$$x(5-5x) = -30$$

$$5x - 5x^2 = -30$$

$$\text{OR } -5x^2 + 5x + 30 = 0$$

This is the standard form of quadratic equation.

FACTORIZING METHOD

$$x^2 - 3x + 2x - 6 = 0$$

$$x(x-3) + 2(x-3) = 0$$

$$(x-3)(x+2) = 0$$

$$(x+2) = 0$$

$$x = -2$$

$$(x-3) = 0$$

$$x = 3$$

$$x = 3$$

$$x = -2$$

Two values of x.

Putting these values of x in equation (ii) and find the value of y.

$$(ii) \quad 5x + 4y = 5$$

$$4y = 5 - 5x$$

$$y = \frac{5-5x}{4}$$

At

$$x = -2$$

$$y = \frac{5-5(-2)}{4}$$

$$= \frac{5+10}{4} = \frac{15}{4} = 3\frac{3}{4}$$

At

$$x = 3$$

$$y = \frac{5-5(3)}{4}$$

$$y = \frac{5-15}{4} = \frac{-10}{4}$$

$$y = \frac{-5}{2} = -2\frac{1}{2}$$

Thus

$$x = -2, \quad y = 3\frac{3}{4}$$

$$x = 3, y = -2\frac{1}{2}$$

EXERCISE - 5

Solve the following simultaneous equations.

$$(1) \quad 2x^2 - 4y^2 = 50, \quad x + y = 6$$

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

- | | | |
|-----------------------------|---|----------------|
| (2) $3x^2 + 2xy + y^2 = 80$ | , | $2x - y = 4$ |
| (3) $x^2 - y^2 = 106$ | , | $2x - 3y = 8$ |
| (4) $3x^2 + 2xy = -8$ | , | $4x = y + 13$ |
| (5) $3x^2 + 2xy + y^2 = 11$ | , | $3x - 4y = 19$ |

ANSWERS

- | | | |
|----------------------|---|-----------------------|
| (1) $x = 5.15$ | , | $y = 0.85$ |
| $x = 18.8$ | , | $y = -12.8$ |
| (2) $x = 3.74$ | , | $y = 3.48$ |
| $x = -1.56$ | , | $y = -7.12$ |
| (3) $x = 11.425$ | , | $y = 4.95$ |
| $x = -17.825$ | , | $y = -14.55$ |
| (4) $x = 2$ | , | $y = -5$ |
| $x = \frac{4}{11}$ | , | $y = -11\frac{6}{11}$ |
| (5) $x = 1$ | , | $y = -4$ |
| $x = 2\frac{23}{81}$ | , | $y = -3\frac{1}{27}$ |

GRAPH OF LINEAR EQUATION

To make the graph of a linear equation, we assign a number of assumed values of independent variable x and find correspondent values of dependent variable y . It provides us pairs of values, which are plotted in the graph.

If we assume values of x $(-3, -2, -1, 0, 1, 2, 3)$

Value of y w.r.t. every value of x may find out suppose $y = 4x - 1$

Is a linear function.

We proceed to make the graph according to above mentioned process.

Schedule

- | | | |
|-------|-------------------|-----------------|
| | $y = 4x - 1$ | |
| (i) | $y = 4(-3) - 1 =$ | $-12 - 1 = -13$ |
| (ii) | $y = 4(-2) - 1 =$ | $-8 - 1 = -9$ |
| (iii) | $y = 4(-1) - 1 =$ | $-4 - 1 = -5$ |
| (iv) | $y = 4(0) - 1 =$ | $0 - 1 = -1$ |
| (v) | $y = 4(1) - 1 =$ | $4 - 1 = 3$ |
| (vi) | $y = 4(2) - 1 =$ | $8 - 1 = 7$ |
| (vii) | $y = 4(3) - 1 =$ | $12 - 1 = 11$ |

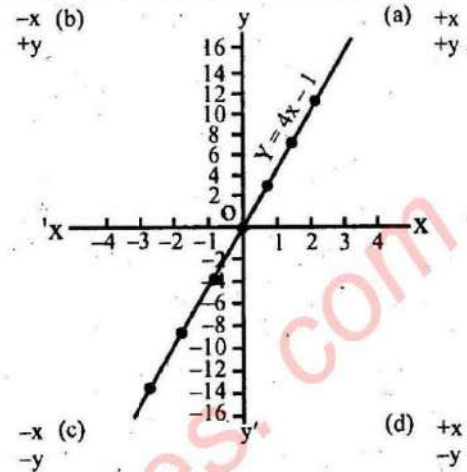
Graph

As it is a Linear Equation, its Graph will be a straight line.

In view of above information we prepare the following schedule showing various pairs of x and y .

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

X	-3	-2	-1	0	1	2	3
Y	-13	-9	-5	-1	3	7	11



GRAPH OF AN ECONOMIC LINEAR FUNCTION

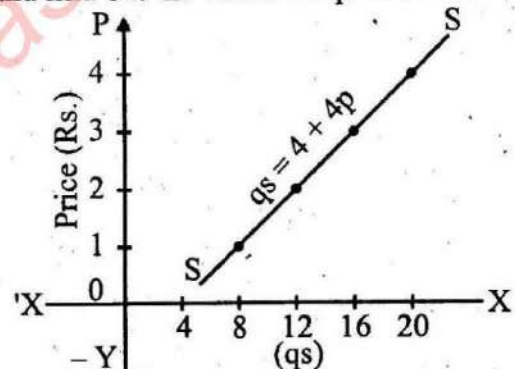
Supply Function: According to the law of supply $q_s = f(p)$ and it is an increasing function. Thus SS is positively sloped.

Suppose $q_s = 4 + 4p$

We give [1,2,3,4] assumed positive values to p and find out the values of q_s as under:

- (i) $q_s = 4 + 4(1) = 4 + 4 = 8$
- (ii) $q_s = 4 + 4(2) = 4 + 8 = 12$
- (iii) $q_s = 4 + 4(3) = 4 + 12 = 16$
- (iv) $q_s = 4 + 4(4) = 4 + 16 = 20$

P	1	2	3	4
qs	8	12	16	20



As supply function is a linear Equation, thus Graph is a straight line.

STATISTICAL CONCEPTS

CONCEPT OF STATISTICS:

Originally the word "Statistics" have been derived from the Latin Word "Status" or Italian word "Statista" or the French word "Statistique", each of which means a political state.

In olden days, statistics was used for the record of Population, Agricultural Product, Govt. Revenue, war heads and the by product of administrative activities of the "state". In the present era of computer, it's scope have been extended in almost every sphere of life. Statistics can be defined in singular and "plural" meanings.

IN singular sence, "statistics is the science of systematic collection, presentation, analysis and interpretation of numerical data to draw conclusions and to make decisions."

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

Technically speaking the word "statistics" is the plural of "statistic" which means the aggregates of numerical facts or data calculated from sample observations in any field for some purpose.

According to Webster, "Statistics means the arranged facts of numerical nature which reveal the specific condition of the people in any field."

A.B. Bouley says, "Statistics is the science of counting." Thus we can say that "Statistics is a science concerned with the collection, presentation, analysis and interpretation of numerical data."

CLASSIFICATION OF STATISTICS

Statistics can be classified into two groups:-

- 1. Statistical Methods.**
- 2. Applied Methods**

1. STATISTICAL METHODS:

The original data regarding any research is very complex and un-arranged. It is brought into simple and clear shape through statistical methods. This method provides us the means and principles which provide guidance with respect to collection presentation, classification, tabulation, comparison and interpretation of numerical data. Future predictions depend upon such data.

2. APPLIED METHODS

When the statistical principles are applied for the analysis of a particular problem, that is known as "Applied Statistics." For instance, the application of specific methods to find out the changes in price level for the last ten years in Pakistan etc. Applied statistics may be qualitative as well as scientific.

STATISTICAL DATA

Statistical data are numerical facts either on counts or measures form relating to some definite purpose under consideration or study.

CLASSIFICATION AND COLLECTION OF DATA

Statistical data may be classified as primary data and secondary data.

1. PRIMARY DATA

These are those "data" collected for the first time and are in "raw form" and "original shape". These have not undergone for any statistical analysis:

METHODS OF COLLECTING PRIMARY DATA

Primary data are generally collected with the help of following methods.

(i) Direct Personal Observation:

Under this method the investigator obtains the relevant and required information himself personally on the spot where inquiry is being conducted.

Information obtained through this method is quite accurate and dependable but the method is slow and expensive.

The investigator must be impartial, well versed and well behaved so that he/she may not get the results of his own choice.

ECONOMICS (EM) NOTES FOR 1st YEAR CLASS (PUNJAB)

(ii) Indirect Personal Observation:

Sometimes the informants hesitate to provide correct and complete information e.g. real income, tax payments etc. Under such a situation information is collected through indirect sources or third parties e.g. evidence of near friends, neighbours, witnesses etc. These sources should be directly or indirectly related with the required facts and must not be biased.

(iii) Mailed, E-Mail and Media Questionnaire Method:

Under this method a "questionnaire" depending upon comprehensive, relevant, simple, brief and clear questions in a standard form is prepared. Blank Questionnaire about the information needed are sent to the informants by Mail, E-Mail and Media, requesting them to return it duly filled as required. This method becomes cheaper and time saving provided the informants are educated, responsible and return it duly filled per requirement.

(iv) Investigation Through Enumerators:

Under this method trained enumerators are sent with the questionnaires to the informants. They try to collect proper and correct answers from relevant persons. This method is costly. In Pakistan this method is used for population census etc.

(v) Through Local Sources:

Under this method information is obtained from agents or local correspondents. This method is cheaper but hardly reliable.

(vi) Through the Record of Registration:

Information may also be collected from the registration of various events e.g. deaths, births, marriages etc. estimation from local union councils and corporations.

2. SECONDARY DATA

The secondary data are the second hand information. These data have already gone through statistical treatment at least once, like collection, classification, tabulation etc. The records or publications which have been already collected for some purposes are available for the users are called secondary sources.

METHODS OF COLLECTING SECONDARY DATA

It is available in two forms in (i) Printed Matter (ii) Non-Printed Matter. The major sources are:

(i) Official Sources:

Statistical data are available from publications of central, provincial and local Governments, the national publications, of statistical division, ministries of finance commerce, production, food, agriculture, health, railway etc.

(ii) Semi-Official Source:

These include the publications of institutions e.g. State Bank, WAPDA, District Councils, Municipalities TCP, PTDC, PIDE, Educational Boards etc.

(iii) Private Sources:

The reports and publications of chambers of Commerce, Trade Markets, Co-operative Societies, Insurance Companies, Banks, N.G.O's etc.

(iv) Publications of Research Organizations:

University Institute of Education and Research, Agricultural PCSIR, Text Book Boards, Punjab Board of Economic enquiry etc.

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(v) Reports of International Organizations:

Such as UNO, UNESCO, UNICEF, WHO, FAO, ILO, IMF, World Bank are reliable sources for collection of data.

(vi) Unpublished data of different departments:

These are available directly from the offices of some departments e.g. Trade Association, LDA, CDA, High Courts, Labour department etc.

EDITING OF DATA

After the collection of data by anyone it becomes necessary to scrutinise the data in order to find out errors, omissions and mistakes. This process is known as "editing of data."

CLASSIFICATION OF DATA

When data in raw form is collected, it's classification is necessary before tabulation, "The process of arranging data into relatively homogenous groups and classes according to their similarities is called classification of data."

The best example of classification is the process of sorting letters in a post office according to Cities, Towns, Streets, Sectors, Villages etc. It may be regarded as 1st stage of Tabulation.

TYPES OF CLASSIFICATION

1. Qualitative Classification:

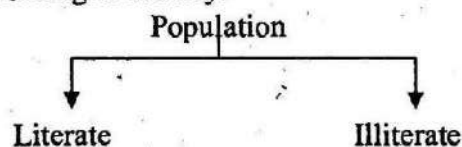
When the data are classified according to some attributes or quality e.g. Religion, Colour, Literacy, Sex, beauty, happiness, health etc. The classification is known qualitative. It can be divided into two types:

(i) Simple Classification:

When the data are classified into only two classes by considering only one attribute such as literate and illiterate, skilled and unskilled etc. the classification is regarded as simple classification.

Example: Classification of population of a city according to literacy.

Total population	Literate	Illiterate
10 lakh	4 lakh	6 lakh

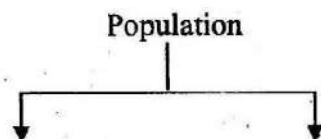


(b) Manifold Classification

When the data are classified on the basis of more than one attributes e.g. Two, Three, Four.... It is termed as manifold classification.

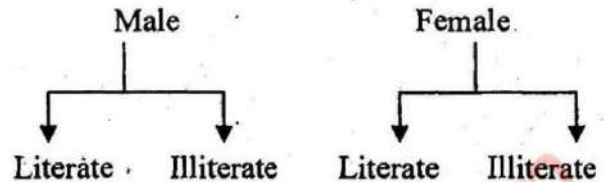
Example: Male:- Literate, Illiterate, Females:- Literate, Illiterate.

Sex	Literate	Illiterate	Total population
Male	2.5 Lakh	2.4 Lakh	4.9 Lakh



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Female	1.5 Lakh	3.6 Lakh	5.1 Lakh
Total	4 Lakh	6.0 Lakh	10.0 Lakh



(b) Quantitative Classification:

If the data are classified according to some quantitative measurement e.g. age weight, height, income, price etc. it is regarded quantitative classification or classification according to class intervals.

It can be subdivided into two parts.

(a) Simple Classification:

It is based on one quantitative characteristic. For Example, we classify 400 students in 1st year in age groups.

Age (years)	Number of students
15 – 17	200
17 – 18	150
18 – 19	50

According to every class intervals there is a lower age limit and an upper age limit. As 200 students lower age limit is 15 years and upper age limit 17 years.

(b) Manifold Classification:

There are more than one quantitative measures in manifold classification e.g. age and grade of 400 students as mentioned below.

Age (years)	Grade A	Grade B	Grade C	Grade D	Total number of students
15 – 17	40	60	80	20	200
17 – 18	8	32	48	32	120
18 – 19	4	16	24	36	80
Aggregate	52	108	152	88	400

TABULATION OF DATA

The process of arranging data into rows and columns in a table is known Tabulation. The purposes of these systematic arrangements of data are comparison and analysis to achieve specific results. The tables may be simple or complex.

TYPES OF TABULATION

1. One-Way Tables

When the data are tabulated according to one characteristic that is known as one-way or simple tabulation.

Following table shows class wise number of students of Govt. College for Women Sargodha.

Class	No. of Students
I year	1000
II year	950

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III year	550
IV year	500
Total	3,000

2. Two way Tables:

Two way tabulation shows that data are tabulated according to two characteristics and provide answers of two questions e.g. class wise number of pass and fail students mentioned as below:

Class	No. of Students passed	No. of Students failed	Total No. of Students
I year	800	200	1000
II year	700	250	950
III year	450	100	550
IV year	350	150	500
Total	2300	700	3000

3. Three Way Tables:

when there are three variable of characteristics which are shown in the same table it is called three way tabulation. It provides information regarding three mutually dependent and inter related questions.

If we divide students in science and arts group then it will be converted into three way table.

Class	Science Group			Arts Group			Aggregate number
	Pass	Fail	Total	Pass	Fail	Total	
I year	325	75	400	450	150	600	1000
II year	300	100	400	400	150	550	950
III year	150	50	200	225	125	350	550
IV year	140	60	200	200	100	300	500
Total	915	285	1200	1275	525	1800	3000

Note: Higher order or Multi ways tables provide answers of many inter-related questions of different variable characteristics.

PRINCIPLES OF TABULATION

Following are some important rules / principles for preparing tables.

1. Tables should be clear and easy to understand.
2. The titles should be in "block letters" and self explanatory.
3. If data are large, more than one table be made.
4. Size of table should be according to paper available.
5. Main heading be limited in number.
6. Big classification of data in thick lines and sub-division in thin lines should be

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shown.

7. Those columns whose data are to be compared should be kept side by side.
8. Units of measurement for every head be clearly mentioned e.g. liter for milk, meter for cloth, kilo grams, metric tons for wheat etc.
9. Estimation of data before Tabulation.
10. For easy comparison different items should be arranged alphabetically or according to size.
11. More important figure or topics should be placed in a box or circle.
12. Rows and columns may be numbered..
13. The items should be date wise or according to Geographical regions / locations.
14. Temporal classification may be according to time in the form of time-series.

ADVANTAGES OF TABULATION

1. Un-arranged and complex data are presented in tables to answer questions about problem under discussion.
2. The whole data in the table are in comprehensive form. Correct results within minimum time period are achieved.
3. Errors and omissions are minimized.
4. Table pave way for further statistical treatment.
5. All the aspects regarding data are lime-lighted.

OBJECTIVE QUESTIONS

Select and tick (✓) Correct answer.

1. Point out the continuous variable
(a) Change in price (b) Change in consumption
(c) Change in speed (d) Change in utility.
2. Which one represents exports.
(a) Y (b) M (c) E (d) X
3. What type of Equation is:-
 $2x^{-3} + 4x^{1/2} + 5 = 8$
(a) Quadratic (b) Irrational (c) Rational (d) Identical
4. $(x + 2)(x - 2) = x^2 - 4$ is an equation:
(a) Conditional (b) Identical (c) Cubic (d) Linear
5. Which one is not a dis-continuous variable
(a) Supply of a good (b) Demand for good
(c) Time (d) Consumption
6. Graph of a quadratic equation is shaped?
(a) Kinked (b) A straight line (c) Parabolic (d) Vertical
7. Which functions two values are inverse to each other.
(a) Decreasing (b) Increasing (c) Implicit (d) Explicit

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8. The process of arranging data in rows and columns is known:
(a) Classification (b) Index Number
(c) Tabulation (d) Measurement
9. Income (y) and Investment (I) are functionally related, what type?
(a) Explicit function (b) Implicit function
(c) Inverse function (d) Decreasing function
10. The quantities which change their values in practical life but are assumed constant for special purposes are known:
(a) Constants (b) Parameters (c) Fixed (d) Variables
11. In Trinomial the number of Terms is
(a) Two (b) Three (c) Four (d) Unlimited
12. $Q = 10 - 3p$ is a function
(a) Increasing (b) Implicit (c) Decreasing (d) Inverse
13. The method under which data are divided into Homogeneous groups is known
(a) Classification (b) Tabulation (c) Arrangement (d) Applied
14. A table which shows more than one attributes of quantities is regarded.
(a) Simple table (b) One-way table
(c) Complex table (d) Strange table
15. What type of variable y is when $y = f(x)$
(a) Dependent (b) Independent (c) Exogenous (d) Parameter

DESCRIPTIVE PART – I

SHORT NOTES

1. What are variables?

Generally the quantities which change their values are known as variables e.g. Income Consumption, Savings, investment, Price, demand, Supply, utility, etc. in economics and Time, temperature, wealth in ordinary life.

2. What is a function?

Some variables are inter-related with each other. When one variable (Y), depends upon other variable (X) then $Y=f(X)$ it can be read as variable Y is a function of variable x. In economics we have numerous variable which have functional relationship e.g. $C=f(Y)$, $S=f(y)$, $q_s=f(P)$, $U=f(Q)$ etc.

3. Bring out difference between an Explicit and Implicit function?

When a variable is totally dependent upon the other variable e.g. $y=f(x)$. Thus Y is an explicit function of x. e.g. $q_d=f(p)$, $q_s=f(p)$ when two variables depend upon each other by giving value to one variable, we get the value of other variable (x or y) is known as an implicit function e.g. $XY=10$, $y=f(I)$, $I=f(y)$ etc.

4. Differentiate between an increasing and decreasing function?

When the values of independent and dependent variables move in the same direction (rising/falling) that is called increasing function e.g. $q_s=f(P)$, $C=f(Q)$, $R=f(Q)$, $C=f(Y)$ etc.

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When the values of independent and dependent variables move in opposite direction is known as decreasing function e.g. $q_d=f(p)$, $MU=f(q)$, $I=f(i)$ etc.

5. Point out the symbols of equations?

Unknowns, Constants, Co-efficient, Index or power. Are regarded as the symbols of an equation.

6. What is the difference between conditional and Identical equation?

Whenever value of the known present in the equation provides the root of equation which cannot be derived under any circumstances will be regarded as an identical equation e.g. $(a+b)^2 = a^2+b^2+2ab$, $(x+y)(x-y) = x^2-y^2$, $S = I$ whereas, in conditional equation the values of unknowns, we get by applying solution can only satisfy the equation. No other value of unknown can satisfy the equation (LHS = RHS)

7. What difference do you feel between rational and irrational equation?

When the power of unknown in an equation is positive and real number is called "Rational equation" e.g. $2x^3+3x^2-x=16$ whereas, if the power of unknown is negative or non-real number is called "Irrational equation" e.g. $x^{-3}+4x^{1/2}+3=0$

8. Write down the formula to solve quadratic Equation?

The quadratic equation may be solved with the help of formula.

$$X = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

9. Write down five principles/ Rules of Tabulation?

- (i) Tables should be clear and easy to understand.
- (ii) The titles should be in block letters.
- (iii) More than one table should be made for large data
- (iv) Size of table should be according to paper.
- (v) Main heading be limited in number.

10. What is tabulation?

The process of arranging data into rows and columns in a table is known as Tabulation.

11. Define "Parameters"?

These are the certain quantities whose values keep on changing in real life but for specific purposes these are assumed to remain the same e.g. $q_s=f(p,c,m,\pi)$ in this function C,M, π are the parameters. Because these are assumed constants. We highlight the law of supply, $q_s = f(P)$

12. Point out the kinds of variables?

- 1) Continuous Variables
- 2) Discontinuous Variables
- 3) Independent Variables
- 4) Dependent Variables

13. Classify an algebraic expression in view of terms. ?

According to "Terms" an algebraic expression is classified as under.
Binomial, Trinomial, polynomial

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14. Find out the methods of collection of primary statistical Data?

- Direct personal observation
- Indirect personal observation
- Mail, E-mail, and media questionnaire method.
- Investigation through enumerators.
- Through local sources
- Through the need of registrations.

15. What are the advantages of Tabulation?

- Un-arranged and complex data are presented in tables to answer questions.
- Presentation of data in comprehensive form for correct result within minimum time period.
- Errors and omission are minimized.
- Table paves way for further statistical treatment.
- All aspects regarding data are time-lighted.

PART – II

1. Define variables and Discuss their kinds.
2. What is an equation. Discuss various kinds of equations.
3. Define the following.
 - (a) Identical equations.
 - (b) Simultaneous equations.
 - (c) Linear equations.
4. Prepare the Graph of following linear equations
 - (i) $y = 8 - 2x$
 - (ii) $q = -p + 12$
5. What is meant by Tabulation. Discuss various types of Tabulation.

